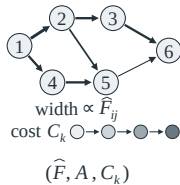


$b_k = B_k / L$
per-step budgets

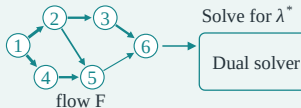
Hard edges H
zero flow

Empirical graph



GAE: A_r, A_c

Joint budget pricing



Solve for λ^*

Dual solver

$$\min_F -\langle A, F \rangle + \varepsilon \text{KL}(F \| \hat{F})$$

$$\text{Flow balance} \mid \langle C_k, F \rangle \leq b_k \mid F_H = 0$$

λ^*

Clip +
smooth

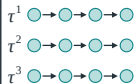
$\bar{\lambda}^*$

Priced policy update

$$\tilde{A} = \frac{A_r - \sum_k \lambda_k A_{c_k}}{1 + \sum_k \lambda_k}$$

PPO / TRPO $\rightarrow \pi_{\text{next}}$

Rollout batch



states, actions
rewards, costs
under policy π

episode budget B

Cost feedback

$$\beta \leftarrow [\beta + \eta(\bar{J} - B)]_+$$

Episode costs $\bar{J}$

+

λ

next rollout